



# Flow boiling heat transfer in two-phase micro-channel heat sinks—I. Experimental investigation and assessment of correlation methods

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## Abstract

This paper is the first of a two-part study concerning measurement and prediction of saturated flow boiling heat transfer in a water-cooled micro-channel heat sink. In this paper, new experimental results are discussed which provide new physical insight into the unique nature of flow boiling in narrow rectangular micro-channels. The micro-channel heat sink contained 21 parallel channels having a  $231 \times 713 \mu\text{m}$  cross-section. Tests were performed with deionized water over a mass velocity range of 135–402  $\text{kg/m}^2 \text{s}$ , inlet temperatures of 30 and 60  $^{\circ}\text{C}$ , and an outlet pressure of 1.17 bar. Results indicate an abrupt transition to annular flow near the point of zero thermodynamic equilibrium quality, and reveal the dominant heat transfer mechanism is forced convective boiling corresponding to annular flow. Contrary to macro-channel trends, the heat transfer coefficient is shown to decrease with increasing thermodynamic equilibrium quality. This unique trend is attributed to appreciable droplet entrainment at the onset of annular flow regime development, and the increase in mass flow rate of the annular film by droplet deposition downstream. Eleven previous empirical correlations are assessed and deemed unable to predict the correct trend of heat transfer coefficient with quality because of the unique nature of flow boiling in micro-channels, and the operating conditions of water-cooled micro-channel heat sinks falling outside the recommended application range for most correlations. Part II of this study will introduce a new annular flow model as an alternative approach to heat transfer coefficient prediction for micro-channels.

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## 1. Introduction

Two-phase micro-channel heat sinks are devices that feature flow boiling of a liquid coolant through parallel channels having a hydraulic diameter of 10–1000  $\mu\text{m}$ . These devices are ideally suited for high-heat-flux dissipation from small surface areas in a broad range of emerging technologies. The combination of small flow passage area and flow boiling produce very high heat

transfer coefficients with minimal flow rate and overall coolant inventory requirements, and provide better stream-wise temperature uniformity than single-phase micro-channel heat sinks [1–6].

Deployment of two-phase micro-channel technology requires a comprehensive fundamental understanding of virtually all hydrodynamic and thermal aspects of phase change in small channels. The ability to accurately predict pressure drop and flow boiling heat transfer for a given micro-channel geometry and operating conditions is of paramount importance to both the design and performance assessment of a micro-channel heat sink.

Because the interest in implementing these devices is fairly recent, brought about mostly by thermal

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